

Jorge Espinoza Abud

Data Scientist

Email: resoluble.io@gmail.com

Telegram: [telegram.me/jorge_abud](https://t.me/jorge_abud)

Linkedin: in/jorgeabud/

Website: jabud.github.io/

About

My professional background has led me to take on multiple roles across various industries:

- I started my career as a production engineer in the marine industry, designing propellers for ships.
- I have worked as a data engineer in the real estate sector, building ETL pipelines and scrapers.
- I've been a software developer at a consulting firm.
- I also worked independently as a Python developer and data scientist on projects in the marketing, cloud computing, and ride-hailing industries.
- I have grown as a data scientist in the fintech industry, leading projects that supported decision-making and had an impact on risk management, customer behavior, and product growth, while also ensuring that processes remained optimized and standardized.

Currently as a Lead Data Scientist at UP.Labs, I am helping grow a new venture-backed startup from the ground up, working on solving major challenges and revolutionizing the transportation industry through data, data science, and artificial intelligence.

Experience

UP.Labs

- **Lead Data Scientist** (Nov 2025 - Present)
 - Helping grow a new startup from the ground up, which secured **~\$3M USD in funding** to operate through its first year.
 - **Leading the data team** and overseeing **analytics, data science, and data engineering** projects.
 - **Designing team roadmaps** and maintaining **constant cross-functional communication** with different teams and functions across the startup.
- **Senior Data Scientist** (May 2025 - Nov 2025)
 - Building data driven startups.

Bankaya

- **Lead Data Scientist** (Jun 2023 - Feb 2025)
 - Led a team of data scientists, providing technical guidance and driving risk and fraud detection initiatives.
 - Designed, trained, and deployed machine learning models for risk and fraud across all company products, **improving bad rates while maintaining approval rates**.
 - **Increased fraud detection related to unwillingness to pay across the portfolio**, enhancing risk assessment and mitigation strategies.
 - Developed an internal **Python library** to streamline the data science workflow, **reducing development time and ensuring standardization**.
 - **Architected the feature store** and built scalable pipelines for **training data generation and real-time feature serving in production**.
 - Established standardized processes and templates for **model training, evaluation, monitoring, and feature engineering**, significantly **reducing time-to-first baseline model and time-to-deployment**.
 - Defined **cross-functional collaboration strategies** with business teams, implemented **best practices for data science operations**, and maintained **comprehensive documentation**.
 - Assessed data providers, conducted **proof-of-concept experiments**, and recommended high-value **data integrations** to enhance model performance and business impact.
 - Designed and evaluated **A/B tests**, performing **power analysis and statistical tests** to determine **sample sizes, experiment duration, and data-driven decisions** for **credit offers and product optimizations**.

Konfio

● **Lead Data Scientist** (Apr 2023 - Jun 2023)

- Mentored data scientists, providing **technical guidance** to refine project details while fostering **independence and skill development**.
- Led **data science projects for collections**, collaborating with stakeholders to **prioritize initiatives**, design **machine learning solutions**, and **communicate updates and results** effectively.

● **Senior Data Scientist** (May 2021 - Apr 2023)

- **Developed early warning detection systems** to help the collections team take proactive action at both **industry and client levels**.
- Built an **income estimation model** to assign credit lines to clients with **incomplete financial data**, improving credit decisioning.
- Designed a **customer lifetime value (CLV) model** to forecast revenue from card clients, supporting **long-term profitability strategies**.
- Developed a **customer churn prediction model**, enabling targeted retention strategies for high-risk clients.
- Created a **propensity model** to estimate customer likelihood of accepting **credit line offers**, optimizing sales team prioritization.
- **Enhanced internal data science tools** to improve efficiency in **model monitoring and debugging**, reducing troubleshooting time.

● **Data Scientist** (Nov 2019 - May 2021)

- **Collaborated across three squads**—Growth, Risk, and Collections—delivering data-driven solutions tailored to each domain.
- Developed a **time series model** to analyze **credit search behavior at the industry level**, identifying trends and market shifts.
- Built a **risk estimation model** for **recurring working capital clients**, improving credit decisioning and risk management.
- Designed a **collection optimization model** to predict the **best date to collect payments from clients**, enhancing recovery rates and operational efficiency.

Self-employed

● **Data Scientist - Python Developer** (Feb 2018 - Nov 2019)

- **@San pancho (now jellyfish)**: Conducted **descriptive data analysis** to identify **customer segments and behavioral patterns**. Developed a **forecasting model** to predict trends in customer behavior, optimizing **targeting strategies**.
- **@Teranalytics**: Designed a **business model for a new product** using **Monte Carlo simulations**. Built a **web-based platform** to visualize results, enabling clients to make **data-driven business decisions** based on future projections.
- **@BlackInnTech**: Developed an **ETL pipeline** and leveraged **NLP models** to process **unstructured text data**, extracting key insights for **background checks and driver hiring** for a major transportation services app.
- **@Sanabria**: Assisted with **web development** and **IT support** tasks.

Kalypso

● **Senior Developer** (Apr 2017 - Feb 2018)

- **Developed software solutions** for the **Product Life-Cycle Management (PLM) consulting team**, enhancing process efficiency and data management.
- **Tech stack**: Java, Oracle platform.

Intelimetrica

● **Lead Data Engineer** (Jul 2016 - Apr 2017)

- Led the **development and maintenance of an ETL system** for **real estate data**, ensuring a reliable data pipeline for company services and analytics.
- Developed **NLP models** to extract **key insights from real estate data**, enhancing **new features** for predictive models in the company.

● **Data Engineer** (Mar 2016 - Jul 2016)

- Developed and maintained **web scrapers**, ensuring **efficient data extraction** for real estate analysis.

Rice Propulsion

- **Production Engineer** (2013 - 2014)
 - **Contributed to the design phase** of the **marine propeller production process**, optimizing efficiency and performance.
 - Worked on the **mechanical design** and performed **stress-strain analysis** for a new product, ensuring structural integrity and reliability.

Education

Cypher Fundamentals

Certificate ID: 990c4cf7-f1de-4a88-a9d4-163bb25ee4ad

Experimental Design Basics

Certificate ID: 92238e4306f0fdf12620664bff4f7f67

A Crash Course in Causality: Inferring Causal Effects from Observational Data

Certificate ID: 48a72e1b6d07e85be20c8f93709c23b1

Perform data science with Azure Databricks

Certificate ID: 2c3d06a1771e6ab15a81159fafbc94d5

Microsoft Azure Databricks for Data Engineering

Certificate ID: bb67deb56f7b56dddb9ed18a6ef23aaa

BSc Mechatronics Engineering

Instituto Tecnológico y de Estudios Superiores de Monterrey, Campus Guadalajara.

Projects

QR generator

I was curious about pyscript and I created a github page with a [QR code generator](#) using python and static website files.

Math animations

Project to visualize and animate math objects and algorithms, using python and matplotlib.

CPMX8 A.I. and Robotics Hackathon

The team and I developed a computer vision model and a simple web application to activate a web-cam, recognize faces and classify them by sentiment. We played with keras, IBM watson and flask.

Sentiment Analysis Classifier

A just for fun and learning purposes project, I used Twitter API to extract text and then build a model to classify it by sentiment.

Skills

Core Competencies

- Leadership
- Analytics & Data Science
- Data Governance
- MLOps
- Proof of Concept

Libraries & frameworks

- Sklearn
- Pandas
- Numpy
- Pyspark
- MLFlow
- statsmodels
- statsforecast

Languages & tools

- Python
- Docker
- Git
- Neo4j
- Databricks
- Claude Code

Languages

Spanish, English